

Data Storage System Vulnerabilities

CompTIA PenTest+ (PT0-003)

Domains:

- 3.0: Vulnerability Discovery and Analysis
- 4.0: Attacks and Exploits

Objectives:

- 3.1: Given a scenario, conduct vulnerability discovery using various techniques
- 4.5: Given a scenario, perform web application attacks using the appropriate tools

Video Overview

Welcome to this video on Data Storage System Vulnerabilities! This video will help you understand the common weaknesses in how data is stored, whether it's on a USB drive, a company server, or in the cloud. As a penetration tester, knowing where and how data is vulnerable is essential for protecting sensitive information in any organization. We'll explore various types of vulnerabilities, from simple misconfigurations like default passwords to more complex issues stemming from outdated software and misleading error messages. By understanding these concepts, you'll be better equipped to identify and address real-world security challenges, ultimately helping businesses safeguard their most valuable asset: their data.

Key Topics Covered

- Data Storage System Vulnerabilities: Overview of common weaknesses in data storage
- Scope of Data Storage Systems: Databases, cloud storage, and physical storage devices
- Sensitive Data Targets: PII, PHI, credit card information, R&D schematics
- Misconfigurations: Default or blank credentials, network exposure, flat networks
- Improper Permissions: "World read/write" settings and their impact
- Logic Flaws and Input Sanitization: Web front-end directly connected to databases, SQL injection

- Firmware and Software Vulnerabilities: Outdated firmware (e.g., QNAP systems)
- Error Handling Weaknesses: Verbose error messages revealing system information

Student Notes Section